zine

source

Haskell is the glue, so that you can think about the parts in isolation. Category theory style Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

This resolves a nominal into a specific noun

$$\mathit{resolve} :: \mathbf{Nominal} \to \mathbf{Noun}$$

These are all the thirty-seven relations in the Universal Dependencies grammar [1].

```haskell
data Relation = Nsubj
              | Obj
              | Iobj
              deriving (Eq, Ord, Enum)

newtype Dependency source target = Dependency (source → ([Relation], target))
```

Because Dependency is generic on any source or target, coreference can be handled separately on a PoS layer I bet

```haskell
–instance Category Dependency where
–  id = Dependency $ \source → ([], source)
–  (Dependency a) . (Dependency b) = Dependency $ \source →
–    let (relations, intermediate) = b source
–        (relations', target) = a intermediate
–    in (relations ++ relations', target)
```

Dependency parsing happens sentence-by-sentence

```haskell
type Syntax = [Sentence]
```

UDPipe (maybe see https://github.com/fpco/inline-c), lazily

$$\mathit{dependencyParse} :: \mathbf{Text} \to \mathbf{Syntax}$$

Functor from the tree-category of Dependency to a meaningful category. See below Section 2.2 – use Prolog

$$\mathit{understand} :: \mathbf{Syntax} \to \mathbf{Maybe\ Semantics}$$

$$\mathit{explain} :: \mathbf{Semantics} \to \mathbf{Syntax}$$

Maxima here, which needs a ECL FFI or general Lisp IPC

$$\mathit{simplify} :: \mathbf{Math} \to \mathbf{Math} \text{ -- Maxima}$$

Yi, The keybindings part

$$\mathit{interpret} :: \mathbf{Input} \to \mathbf{Command}$$

Yi, stuff from EditorM for example

$$\mathit{execute} :: \mathbf{Command} \to \mathbf{State} \to \mathbf{State}$$

Custom, using typst-layout, which needs rust FFI or some (??) IPC

$$\mathit{view} :: \mathbf{State} \to \mathbf{DisplayList}$$

WebRender ofc, which needs rust FFI (see https://github.com/harpocrates/inline-rust) or general Rust IPC

$$\mathit{webrender} :: \mathbf{DisplayList} \to \mathbf{GL}$$

$$\mathit{createFrame} :: \mathbf{IO\ GLContext}$$

$$\mathit{read} :: \mathbf{IO\ Input}$$

The main interaction loop

```
main :: IO ()
main = do
  frame ← createFrame
  let interactionLoop state = do
        display ((webrender . view) state) frame
        input ← read
        let state' = (execute . interpret) input state
        interactionLoop state'
  interactionLoop initialState
```

Profane definitions of types

```
newtype GL = GL { display :: GLContext → IO () }
```

Haskell default and glue:
- Prolog (NLP Topos FOL). Maybe this can be in native Haskell or scheme
- Lisp (CAS) Better interop with scheme
- Rust (rendering)
- C++ (text tagging, Firefox IPDL) Can be done with udpipe-rs

lol maybe idris instead of haskell -> scheme -> rust marwood or other scheme implementation for display list, udpipe-rs -> prolog with call/cc -> lisp direct interop with scheme lists but embedded via ecl2

> if they're all separate, you could get a prolog repl, an ECL repl, and a GHCI repl
>
> maybe prolog translates directly into lispy things
>
> haskell -> prolog -> lisp

1. Philosophy and Sheaf Theory

Need to think about the pushout of rooted words, which would connect subjects to objects via some actual relation. Because of the categorical structure of Dependency, only the dependency paths between two different tokens matters. Pushouts can be related to specific morphisms, and will probably be through things like verbs

If I was to use sheaf theory here, then I've broken the problem into:

- Local semantics
- gluing, including coreference probably
- Global semantics

Globally, I want to extract stuff like

```haskell
newtype Epistemology = Epistemology (Set Fact -> Knowledge)
```

What is true about both the local semantics and the global semantics? Both subcategories of Hask, likely, as they both need to be representable internally by the computer ultimately.

So what is Epistemology? It's a subcategory with only two objects and it goes unidirectionally. pushouts are what two epistemic systems have in common and pullbacks are the dual

what about hegel

```haskell
dialectic :: (Thesis, Thesis) -> Maybe Thesis
```

metaphysics: what is? epistemology: what is knowledge? ethics: what is just/right/needful?

probably can't constrain this too much – maybe it's literally just collections of morphisms between two Hask objects

```haskell
type Philosophy a b = [a -> b]
```

ok now we might be getting somewhere, as I can also make morphisms from local semantics via pushouts through a tree root in the dependency parse. I don't necessarily have to ask from whence nouns came from (this is restricted to humans), just the relationships defined between them.

2. Category Theory

2.1. Definitions

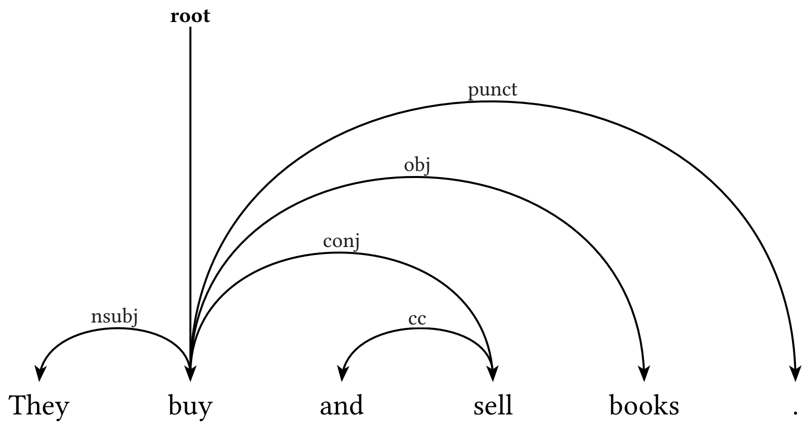
If you're already familiar with categories and their representation in Haskell, you can skip this section. (There is already Control.Category, but it is necessarily a *wide* category, where its objects are always Ob(**Hask**).)

Haskell categories

```haskell
class Category (k :: Type -> Type -> Type) where
  type Object k (a :: Type) :: Constraint

  id :: Object k a => k a a

  (.) :: (Object k a, Object k b, Object k c) => k b c -> k a b -> k a c

instance Category (->) where
  type Object (->) a = ()
  id x = x
  (g . f) x = g (f x)
```

2.2. A Dependency Tree Topos

Consider a dependency tree



To any dependency parse, we can associate a category, say $\mathcal{D}$, by considering the objects as words (and their indices) and the morphisms as dependencies. To be precise, $\mathrm{Ob}(\mathcal{D}) = \{(\mathrm{They}, 1), (\mathrm{buy}, 2), ...\}$ and $\mathrm{mor}(\mathcal{D}) = \{((\mathrm{buy}, 2), [nsubj], (\mathrm{They}, 1)), ...\}$. I'll use $\mathrm{buy}_2 \xrightarrow{\mathrm{nsubj}} \mathrm{They}_2$ as notation for $\{((\mathrm{buy}, 2), nsubj, (\mathrm{They}, 1))\} \in \mathrm{mor}(\mathcal{D})$ from here forward.

Topos of local truth constructed in just one sentence. First add necessary pushouts of morphisms through relation words to construct just noun phrases as objects in the category. Objects are nominals that are related to

The category of elementary topoi $\mathbf{Top}$ is a 2-subcategory of the 2-category $\mathbf{Cat}$

Maybe– Terminal object: "this"/unrestricted PRON Binary products: different sentences, indicates that two independent clauses can be broken up Equalizer: CCONJ/union Exponential: linguistic recursion/which Omega: "itself" all nouns Sub object classifier: X–copula–>"itself"

Existential quantification would use "this", whereas the default is just universal. i.e. Red is bright vs this red is bright.

Now that I have a sheaf topos, I would like to be able to glue it together into a general understanding. The glue will inevitably have a nonzero cohomology in which case we will fail to synthesize a global understanding. But say we have a global understanding; then we can understand how different parts fit together by alternating between natural language representation and the categorical.

Once I have a local topos, I can attempt gluing using some kind of ACL2 combo or H-M unification, or both. I will need to figure this out later, and the exact system by which the local, overfit, understanding gets generalized to the global will determine the cohomology.

Maybe have a two-level sheaf attempting to glue sentences together in one argument, using some natural deduction and coreference (H-M unification there?) in the glue between sentences which are part of a topos. Then, can try a different kind of glue to see how two arguments fit together, and which way the functors go between them. Maybe the higher level isn't even a sheaf and just a topology defined on it.

3. Diagrams

3.1. 3D

Appendix A. Referenced packages

The following imports were used in this Literate Haskell source file.

```haskell
import Data.Kind (Constraint, Type)
import Prelude hiding (id, (.), read)
```

Appendix B. Stubs

```haskell
resolve = undefined
dependencyParse = undefined
understand = undefined
explain = undefined
simplify = undefined
interpret = undefined
execute = undefined
view = undefined
webrender = undefined
createFrame = undefined
read = undefined
initialState = undefined
dialectic = undefined

data Nominal = Nominal
data Noun = Noun
data Sentence = Sentence
data Text = Text
data Semantics = Semantics
data Math = Math
data Input = Input
data Command = Command
data State = State
data DisplayList = DisplayList
data GLContext = GLContext
data Set a = Set
data Fact = Fact
data Knowledge = Knowledge
data Thesis = Thesis
```

Appendix B. References

[1] M.-C. De Marneffe, C. D. Manning, J. Nivre, and D. Zeman, "Universal dependencies," *Computational linguistics*, vol. 47, no. 2, pp. 255–308, 2021.